

Experiment-21

Aim

To perform image sharpening using a Laplacian mask with an extension of diagonal neighbours and enhance the edges and fine details of the image.

Procedure

1. Select and load the input image.
2. Convert the image into grayscale.
3. Define the Laplacian mask with diagonal neighbours.
4. Apply the mask to the grayscale image using convolution.
5. The Laplacian operation detects rapid intensity changes and extracts edges.
6. Since the center coefficient is negative, subtract the Laplacian image from the original image to obtain the sharpened image.
7. Clip the resulting pixel values to the range 0–255.
8. Display the original, Laplacian, and sharpened images.
9. Observe the enhancement of edges and fine details.

Python Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
mask = np.array([
    [1, 1, 1],
    [1, -8, 1],
    [1, 1, 1]
], dtype=np.float32)
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)
```

```
sharpened = gray.astype(np.float32) - laplacian
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)
laplacian_display = cv2.convertScaleAbs(laplacian)
cv2.imshow("Original Image", gray)
cv2.imshow("Laplacian Image", laplacian_display)
cv2.imshow("Sharpened Image", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The image was successfully sharpened using the Laplacian mask with an extension of diagonal neighbours. The mask detects edges in all directions, including diagonal edges, resulting in enhanced edges and fine details in the image.

